identityManagement

Service Description

Service ID: *"identityManagement"*

Abstract
This document describes the `identityManagement` service, version 1.0.0, produced by the Authentication core system of Arrowhead 5.2.2.

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1 Overview

This document describes the `identityManagement` service, which provides operator management of the identities known to Authentication (systems, their credentials and sysop flag) and of their active sessions. It is produced by the Authentication system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

New in Arrowhead 5.

1.2 How This Service Is Meant to Be Used

Operators and the onboarding procedure create identities before a system's first login, update credentials or the sysop flag, query identities and sessions, and close sessions of misbehaving systems.

1.3 Important Delimitations and Dependencies

Access is governed by the management policy of Authentication (`management.policy=authorization` by default: ConsumerAuthorization decides). In this workspace only `Sysop` is created as a sysop identity (finding J23). Passwords are stored as hashes and never returned.

2 Service Interface

This section describes the interfaces to the `identityManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

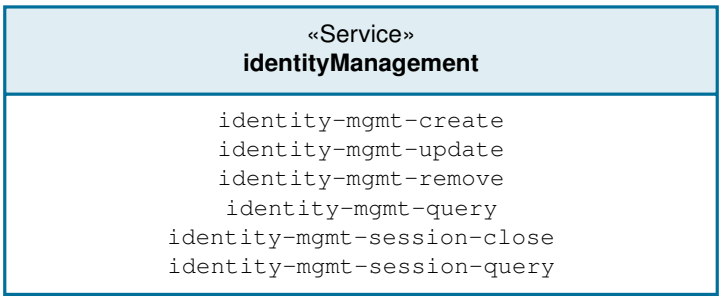


Figure 1: SysML-style block of the `identityManagement` service and its operations.

The following interface operations are available.

2.1 operation identity-mgmt-create (IdentityListMgmtCreateRequest) : IdentityListMgmtResponse

Creates identities with credentials of the given authentication method.

2.2 operation identity-mgmt-update (IdentityListMgmtUpdateRequest) : IdentityListMgmtResponse

Updates credentials and sysop flag of identities.

2.3 operation identity-mgmt-remove (List<Name>)

Removes identities by system name, closing their sessions.

2.4 operation identity-mgmt-query (IdentityQueryRequest) : IdentityListMgmtResponse

Returns a page of identities matching the filter.

2.5 operation identity-mgmt-session-close (List<Name>)

Closes the sessions of the named systems.

2.6 operation identity-mgmt-session-query (IdentitySessionQueryRequest) : IdentitySessionListMgmtResponse

Returns a page of active sessions.

3 Information Model

Here, all data objects that can be part of the `identityManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.11, which are used to represent things like hashes and identifiers.

3.1 struct IdentityListMgmtCreateRequest

Identities to create.

Field	Type	Description
authenticationMethod	String	Authentication method of the identities; `PASSWORD` is the only implemented method.
identities	List<IdentityMgmtRequest>	Identities to create, update or return.

3.2 struct IdentityListMgmtResponse

List of identities.

Field	Type	Description
identities	List<IdentityMgmtResponse>	Identities to create, update or return.
count	Number	Total number of matching records (all pages).

3.3 struct IdentityListMgmtUpdateRequest

Identities to update.

Field	Type	Description
identities	List<IdentityMgmtRequest>	Identities to create, update or return.

3.4 struct IdentityMgmtRequest

One identity.

Field	Type	Description
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
credentials	Map<String, String>	Credentials of the authentication method, for `PASSWORD` the key `password`.
sysop	Boolean	Whether the identity is a system operator (default false).

3.5 struct IdentityMgmtResponse

A stored identity (without credentials).

Field	Type	Description
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
authenticationMethod	String	Authentication method of the identities; `PASSWORD` is the only implemented method.
sysop	Boolean	Whether the identity is a system operator (bypasses the management filters).
createdBy	Name	Name of the system that created the record.
createdAt	DateTime	UTC time of creation.
updatedBy	Name	Name of the system that changed the record last.
updatedAt	DateTime	UTC time of the last change.

3.6 struct IdentityQueryRequest

Paged filter for identities.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
namePart	String	Part of the system name to match.
isSysop	Boolean	Only sysop (true) or non-sysop (false) identities.
createdBy	Name	Name of the system that created the record.
creationFrom	DateTime	Lower bound of the creation time.
creationTo	DateTime	Upper bound of the creation time.
hasSession	Boolean	Only identities with (true) or without (false) an active session.

3.7 struct IdentitySessionListMgmtResponse

List of active sessions.

Field	Type	Description
sessions	List<IdentitySessionResponse>	Active sessions.
count	Number	Total number of matching records (all pages).

3.8 struct IdentitySessionQueryRequest

Paged filter for active sessions.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
namePart	String	Part of the system name to match.
loginFrom	DateTime	Lower bound of the login time.
loginTo	DateTime	Upper bound of the login time.

3.9 struct IdentitySessionResponse

An active session.

Field	Type	Description
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
loginTime	DateTime	UTC time of the login.
expirationTime	DateTime	UTC time when the identity token expires.

3.10 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.11 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “Authentication – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, authentication/documentation/v5.2.2/Authentication-v5.2.2-SysD.pdf.

[2] ——, “identityManagement HTTP/TLS/JSON – Interface Design Description (Authentication),” Eclipse Arrowhead, IDD 5.2.2, 2026, authentication-v5.2.2-identityManagement-HTTP-IDD.pdf.

[3] ——, “identityManagement MQTT/TLS/JSON – Interface Design Description (Authentication),” Eclipse Arrowhead, IDD 5.2.2, 2026, authentication-v5.2.2-identityManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	